G4 AI MASTER CHALLENGE 001 / BUSINESS DIAGNOSIS

RavenStack - Beyond the Churn Headline

A monthly, evidence-driven diagnosis of growth, retention and commercial movement

CONDITIONAL DATE RECONSTRUCTION

More events do not automatically mean more customers lost. In the data, event and plan-loss intensity rise while complete account-coverage-loss intensity falls.

Measure the movement. Then decide what to fix.

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Independent submission. No G4 affiliation or endorsement implied.

OBSERVE DECOMPOSE PERSIST BUILD

The record is mixed - and actionable

Events and plan losses intensified; complete-account coverage-loss intensity did not. That distinction changes the decision.

THE BUSINESS FINDING

In 2024 the event log and conditional plan-proxy loss rate rose, but the normalized rate of complete account coverage-loss episodes fell. Fifty-four of 62 plan losses left another plan active; all eight complete-loss episodes later returned. This is not a verified customer-churn trend.

Q1 / WHY?

Diagnosis: 62 reconstructed account-plan losses; in 54 (87.1%), another plan relation remained active, so most were not complete account-coverage losses. Priority hypotheses: features 114/600 (19.0%), support 104/600 (17.3%) and budget 104/600 (17.3%). These are declared reasons, not proof of why customers ended a commercial relationship. Confidence: high on counts; low on causality.

Q2 / WHO?

Enterprise has 5/120 conditional 2024 opening relations lost (4.17%); Basic 2/132 (1.52%); Pro 0/122. Eight complete-loss/return histories merit reconciliation, not risk scoring. Confidence: moderate on date-reconstructed history; low on future risk.

Q3 / WHAT NEXT?

Fix lifecycle/event links; reconcile 60 historical affected accounts; audit tier flags; then randomize eligible accounts into outreach and holdout. Scenario net value ranges from -$1,056 to +$16,080 under stated assumptions. Confidence: high on arithmetic; low on assumptions.

DECISION

Fund measurement and focused case review now. Do not fund broad discounting on the strength of event counts alone.

Scale, rate and unit tell different stories

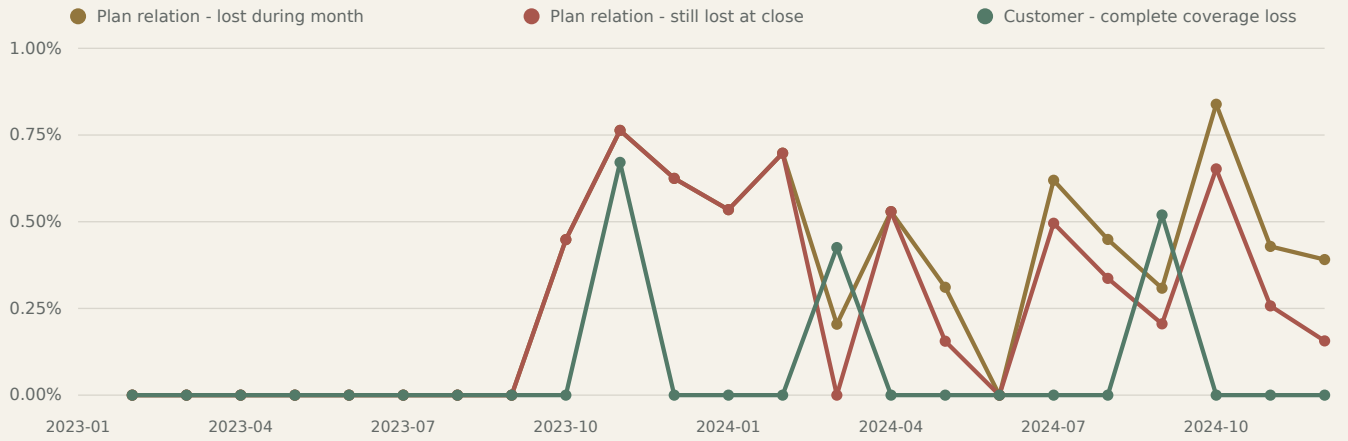
Monthly values aggregate the daily account x plan-tier ledger. Jan-23 has no opening base, so its churn rate is UNDEFINED, not 0%.

Measure	2023	2024	Change / read
Recorded event rows	73	527	+454 rows; events, not lost customers
Opening account exposure	845 acct-months	3,909 acct-months	4.63x exposure
Events / 100 account-months	8.64	13.48	+56.1% recorded intensity
Plan-proxy losses / relation-months	6 / 1,406	56 / 9,414	0.427 to 0.595 per 100
Complete coverage-loss episodes	2	6	0.237 to 0.153 per 100 account-months

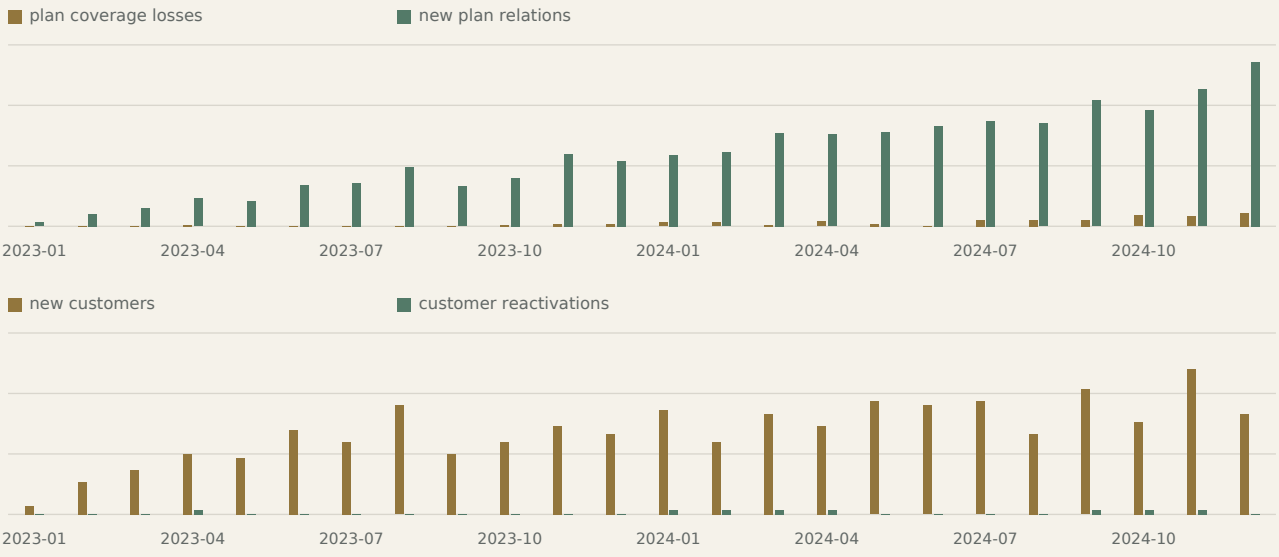
Opening-cohort monthly loss rates (%) - 24 months

OCT '24 · 9 opening-cohort losses / 1,073 = 0.84% rate peak

DEC '24 · 11 gross events; 5/1,279 opening cohort = 0.39%



Monthly flows (counts) - losses vs. additions



The +454 event-row increase decomposes symmetrically into 338.9 rows from greater exposure and 115.1 from higher event intensity. This is arithmetic, not causal attribution; logging change and composition effects remain unidentifiable.

Lifecycle data are the decision gate

Five tables describe different parts of the customer journey. Their joins and time fields limit causal interpretation.

DECLARED EVENT REASONS / N=600

Reason	Rows	Share
Features	114	19.0%
Support	104	17.3%
Budget	104	17.3%
Unknown	95	15.8%
Competitor	92	15.3%
Pricing	91	15.2%

Feature/value, service and affordability deserve separate validation. The reason code is not a verified cause or a unique customer count.

USAGE / SOURCE VOLUME

Usage count: 124,561 to 125,964 (+1.13%). Per observed account-month: 24.462 to 24.747 (+1.17%).

But 19,142/25,000 usage rows predate linked subscription start, and 290 follow recorded end. Only 5,568 rows are interval-valid; not a verified pre-loss window.

SUPPORT / ANY EVENT VS NONE

Measure	Event	No event
Mean satisfaction	3.9804 (815 scored)	3.9833 (360)
Missing scores	580/1395	245/605
Median first response	89.0 min	87.0 min
Median resolution	35.0 h	35.0 h
Account mean escalation	4.93%	4.49%

Small retrospective differences; tickets cluster by account and are not aligned to loss dates. Satisfaction is missing on 825/2,000 tickets.

LIFECYCLE / STRONGEST ACTIONABLE FINDING

500 accounts; 5,000 subscriptions; 600 event rows across 352 accounts. Account churn flags mark 110 accounts; subscription flags touch 312. Account flag vs. any-event history differs for 312/500 accounts.

The subscription table contains 529 upgrade and 218 downgrade flags (23 both), but no dated from/to linkage. Instrument the lifecycle before treating these as migrations.

Prioritize investigation, not prediction

Small, conditional opening-cohort losses identify where to look first - not who is likely to leave next.

Plan-tier proxy	Opening	Lost any time	Rate (95% Wilson)	Still lost
Enterprise	120	5	4.17% (1.79-9.38)	2
Basic	132	2	1.52% (0.42-5.36)	1
Pro	122	0	0.00% (0.00-3.05)	0

Enterprise is 2.65 percentage points above Basic (4.17% vs. 1.52%), but the Wilson intervals overlap widely. No stable tier ranking or calibrated risk follows from 5, 2 and 0 events.

EIGHT COMPLETE-COVERAGE-LOSS EPISODES THAT LATER RETURNED

Account	Segment / source	First loss -> return	Coverage around event
A-432483	Cybersecurity / US / ads	2023-04-05 -> 2023-04-11	Pro lost; Basic
A-180abf	DevTools / US / event	2023-11-06 -> 2024-01-17	Enterprise lost; Basic
A-40a557	Cybersecurity / AU / partner	2024-01-16 -> 2024-02-12	Enterprise lost; Enterprise
A-8adedc	EdTech / US / event	2024-03-09 -> 2024-03-16	Enterprise lost; Basic
A-c9bb3e	EdTech / US / ads	2024-04-02 -> 2024-04-25	Pro lost; Pro
A-0baac2	FinTech / DE / organic	2024-09-13 -> 2024-10-14	Enterprise lost; Basic
A-0a282f	DevTools / US / organic	2024-09-25 -> 2024-09-28	Enterprise lost; Enterprise
A-410e86	FinTech / US / organic	2024-11-22 -> 2024-11-29	Enterprise lost; Enterprise

Why these cases: each has an observed date-derived interval with no plan-tier coverage, followed by a recorded return. Review the eight first, then the broader 60-account loss population. Validate current status before outreach; account-only event reasons cannot be attached to a particular plan.

Measure first. Test retention next.

Five actions translate the strongest evidence into reversible decisions. Only the scenario values are hypothetical.

01 **Instrument lifecycle identity** — Data / RevOps · days 0-30

Link cancellation, reactivation and plan-change events to account + subscription + effective date. KPI: linked and reconciled event share.

02 **Reconcile affected histories** — CS Operations / RevOps · days 0-30

Review 60 historical accounts, starting with 8 complete-loss/return cases. Planning effort: 19 analyst/CS-ops hours (assumed).

03 **Audit Enterprise and tier flags** — RevOps · days 30-60

Test 5/120 Enterprise versus 2/132 Basic and 0/122 Pro against contracts; verify the 529/218 row flags.

04 **Validate stated reasons** — CS / Product · days 30-60

Follow up feature, support and budget reasons separately; link customer-confirmed reason to a lifecycle outcome.

05 **Randomized retention pilot** — CS / Analytics · days 60-90

After eligibility is reliable, randomize at account level; 90-day verified coverage primary, downgrade and support guardrails.

PILOT SENSITIVITY / ILLUSTRATIVE USD-EQUIVALENT / NOT OBSERVED REVENUE OR CAUSAL ROI

Case	Uptake x lift	Retained eq.	Value x months	Gross	Cost	Net	Break-even / mo
Conservative	40% x 2 pp	0.48	$100 x 3	$144	$1,200	-$1,056	$833.33
Base	60% x 5 pp	1.80	$200 x 6	$2,160	$1,200	$960	$111.11
Optimistic	80% x 10 pp	4.80	$400 x 9	$17,280	$1,200	$16,080	$27.78

The 60 accounts are historical investigation cases, not a verified current eligible retention population. All scenarios assume eligibility and hypothetical values; they are not observed revenue or causal ROI.

Model inputs: 60 hypothetical eligible accounts; $20 contact cost each ($1,200 total); scenario uptake 40/60/80%; assumed incremental retention lift 2/5/10 percentage points; monthly value $100/$200/$400; benefit horizon 3/6/9 months. Net = modeled value minus outreach cost only. Conservative case is -$1,056. All inputs require validation; no present-day eligibility, margin, labor, discounting or accounting revenue is observed.

A decision sequence with explicit gates

The paper is the primary deliverable; the local monthly laboratory is a companion for inspecting evidence and assumptions.

WHEN	DELIVER	EXIT GATE
0-30 days	Lifecycle data contract; review eight full-loss/return cases and 60 plan-loss histories.	Effective dates and a customer-level active/lost definition approved by RevOps/CS.
31-60 days	Audit Enterprise cohort and plan flags; validate stated feature/support/budget reasons; instrument activation.	Outcome/event linkage and current eligible population are measurable.
61-90 days	Account-randomized outreach/holdout pilot; publish 90-day effect and guardrails with uncertainty.	Scale only if customer outcome improves and downside guardrails hold.

WHAT THE NUMBERS MEAN

Customer churn rate = unique eligible customers lost in the period / unique active customers at its opening. A commercially verified customer churn KPI is not identifiable here. The conditional 2024 account x plan-tier opening cohort gives 7/374 (1.87%) losing coverage at any time, 3/374 (0.80%) still uncovered at year-end, and 0/190 opening accounts with complete coverage loss. Use this only as a PLAN-LEVEL PROXY under date intervals [start_date, end_date). There is no product_id; dates do not establish effective commercial end; account-only events have no subscription link. Recorded-event intensity is rows per opening account-month, not a churn rate. Recurring revenue changes, causal drivers and calibrated future risk remain UNKNOWN.

REPRODUCE / EXPLORE

Paper source, generator and numerical evidence manifest live in docs/paper/. From solution/: run the documented analysis commands, then `docker compose up --build` for the companion monthly lab. Source hashes and formulas are in the manifest. Dataset credit: River @ Rivalytics (synthetic, MIT-like terms). This is independent of G4; no endorsement is implied.

AUTHORIAL MARK

Coffee - observe. Cube - decompose. Training - persist. Code - test and build. A personal working signature, not a credential claim.